Sahaj Bhadja

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Summary

Early-career software engineer with 2+ years building storage-backed backend systems, cloud data pipelines, and high-throughput APIs across Azure and AWS. Strong in Java, Python, Linux fundamentals, CI/CD, and reliable distributed services.

Education

University of Southern California, Los Angeles, CA **August 2024 -- May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 -- May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 -- Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, tuning indexed queries and access paths to cut dashboard latency from 1.2 seconds to 350 milliseconds for data-heavy workflows.
- Built fault-tolerant background workflows for 1,000+ monthly jobs using explicit state transitions, retry handling, and failure isolation, improving processing throughput by 3x while strengthening reliability across concurrent platform operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 -- August 2025*
- Engineered Flask services, REST APIs, and instructor-facing React workflows handling 10,000+ daily requests, integrating secure file ingestion and automated grading pipelines while improving backend reliability for production educational workloads.
- Designed concurrent batch processing for 1,000+ uploaded files using multithreading and AWS RDS-backed services, reducing grading turnaround time by 60% and strengthening throughput under peak ingestion and evaluation volumes.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 -- May 2025*
- Built Python evaluation pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that improved experiment traceability and strengthened debugging for algorithmic decision and model-analysis workflows.
- Documented experiment behavior, failure cases, and reporting outputs for researchers, turning ad hoc testing into repeatable processes with clearer traceability, faster handoffs, and more reliable benchmarking across research iterations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 -- July 2024*
- Engineered API and ETL microservices using Python, SQL, AWS Glue, and S3, improving response times by 30% while supporting resilient cross-platform data movement, validation, and automation across production workflows.
- Maintained Jenkins-based CI pipelines and 20+ automated test suites with PyTest and Mockito, increasing release reliability by 68% and reducing regressions across Windows, Linux, and macOS service environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 -- August 2023*
- Built REST API and WebSocket features supporting 2M+ synchronized records across workflow modules, improving data consistency and real-time responsiveness for product operations that depended on cross-service coordination.
- Automated Docker-based build and test workflows with Jenkins CI/CD, cutting release time by 50% while improving deployment consistency, troubleshooting speed, and shared validation across developer and staging environments.

Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, C++, Bash
Systems: Linux, computer architecture, multithreading, concurrency, distributed systems, background workers
Backend & APIs: REST APIs, FastAPI, Flask, Node.js, WebSockets, microservices, ETL pipelines
Data & Storage: Azure SQL, PostgreSQL, MySQL, SQL Server, MongoDB, Blob Storage, S3, relational schema design, indexing, query tuning
Cloud & Delivery: AWS, Azure, Docker, Jenkins, CI/CD, Git, automated testing, PyTest, Mockito, reliability engineering

Projects

CPU-scheduling Simulator **Live** | *C++, React, OS Scheduling*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ in C++ with React-based controls and Gantt visualization, enabling metric calculation and timing analysis across 1,000+ simulated process combinations.

Chat2SQL **Code** | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript application with Docker-backed local development workflows.

Big-Brain **Live** | *GCP, FAISS, PostgreSQL*
- Built a secure document platform with vector retrieval and cloud-backed indexing over 50K+ documents, enabling semantic search and summarization while reducing review time by 60% through scalable ingestion workflows.